Sequence 7

ONE-VARIABLE DESCRIPTIVE STATISTICS

7–18 December 2026

I Frequencies, relative frequencies and mean

Definitions

A data set describes a characteristic over a population. If the value x_i appears n_i times, n_i is its frequency count. The nonzero total count is $N = \sum_{i=1}^{p} n_i$; its relative frequency is $f_i = n_i/N$. The mean is

$$\overline{x} = \frac{\sum_{i=1}^{p} n_i x_i}{N} = \sum_{i=1}^{p} f_i x_i.$$

The range is $x_{\max} - x_{\min}$.

Example 7.1: Shoe sizes – booklet data

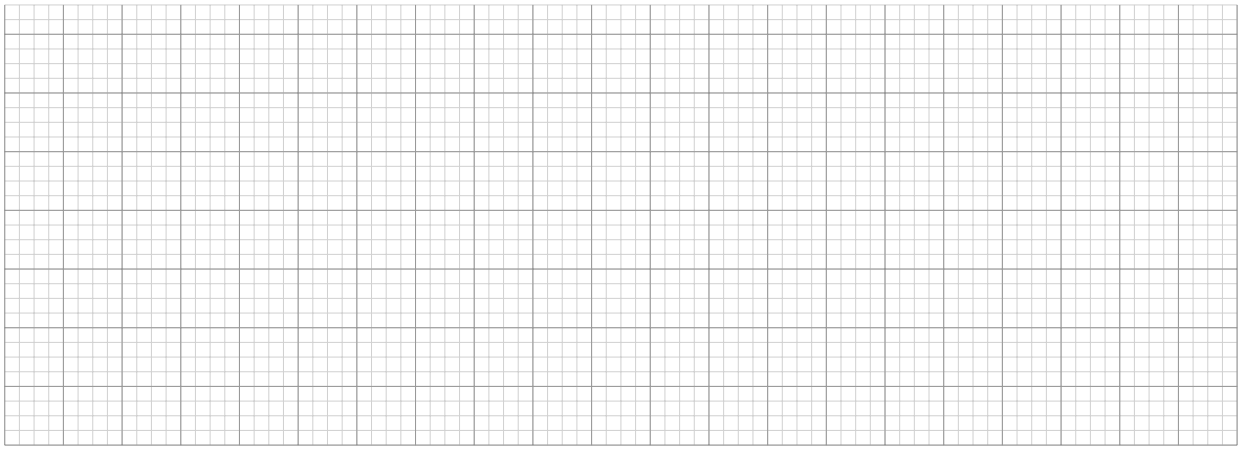
We study the shoe sizes of 30 students:

36	42	40	41	41	38	39	43	42	42	42	40	36	40	39
41	40	38	44	42	40	42	43	38	43	36	41	41	44	43

Complete the counts, calculate the relative frequencies, the mean and the range.

Size	36	37	38	39	40	41	42	43	44
Count	.....	.....	.....	.....	.....	.....	.....	.....	.....

Solution:



II Median and cumulative counts

Definition and convention

A median is a number such that at least half of the values are less than or equal to it, and at least half are greater than or equal to it. In the ordered data set of N values, if N is odd, take the value of rank $(N+1)/2$; if N is even, choose the mean of the values of ranks $N/2$ and $N/2+1$.

Watch out. With repeated values, more than 50 % of the values can be less than or equal to the median.

Example 7.2

Complete the ascending cumulative counts (ACC) of the shoe sizes and determine the median.

Size	36	37	38	39	40	41	42	43	44
Count	3	0	3	2	5	5	6	4	2
ACC	.....	.....	.....	.....	.....	.....	.....	.....	.....

Solution:

Example 7.3

Calculate the medians of $1;2;2;8;9$ and of $1;2;8;9$.

Solution:

III Quartiles and box plot

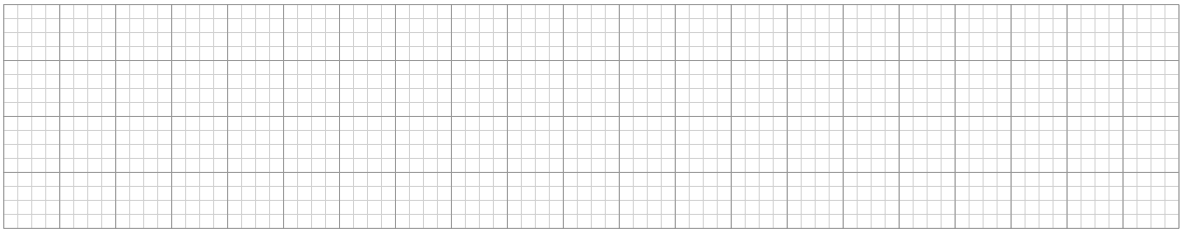
Definitions

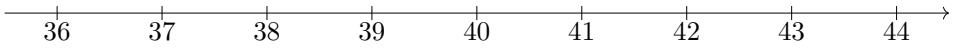
The first quartile Q_1 is the value of rank $\lceil N/4 \rceil$: the smallest value of the data set such that at least 25 % of the values are less than or equal to it. Q_3 is the value of rank $\lceil 3N/4 \rceil$. The interquartile range is $Q_3 - Q_1$. The box plot shows the minimum, Q_1, the median, Q_3 and the maximum.

Example 7.4

Determine Q_1, Q_3 and the interquartile range of the shoe sizes, then draw the box plot.

Solution:

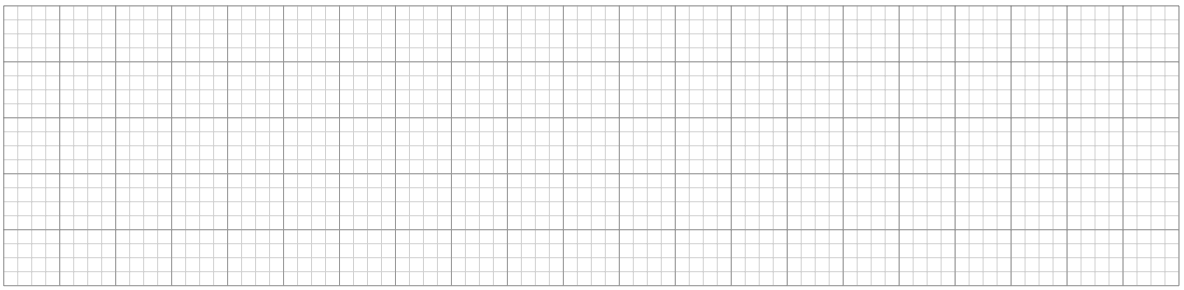




Example 7.5

Determine the quartiles of $1; 2; 2; 3; 4; 4; 4; 5$. Are exactly 25 % of the values less than or equal to Q_1?

Solution:



IV Adding or removing a value

Properties

Adding a to a data set of size N gives $\overline{x}' = (N\overline{x} + a)/(N+1)$. Removing one occurrence of a, for $N > 1$, gives $\overline{x}' = (N\overline{x} - a)/(N-1)$. Use the exact sum, with no intermediate rounding. Re-examine the median in the new ordered data set.

Example 7.6

In the shoe-size data, the sum is 1217 and the median 41.

1. Add a size 36 and calculate the new statistics.
2. Starting again from the initial data, remove a size 44 and repeat.

Solution:

Example 7.7

Justify that adding the mean does not change the mean. Can the range decrease when a value is added?

Solution:

V Variance and standard deviation

Definitions

$V = \dfrac{\sum n_i(x_i - \overline{x})^2}{N}$ is the variance; $\sigma = \sqrt{V}$ is the standard deviation. It measures the spread around the mean and has the same unit as the data. One can also compute $V = \dfrac{\sum n_i x_i^2}{N} - \overline{x}^2$.

Example 7.8

Calculate the variance and the standard deviation of the shoe sizes.

Solution:

Example 7.9

Compare A : $8; 10; 12$ and B : $0; 10; 20$ using the mean–standard deviation pair and the median–interquartile range pair.

Solution:

VI Linearity of the mean

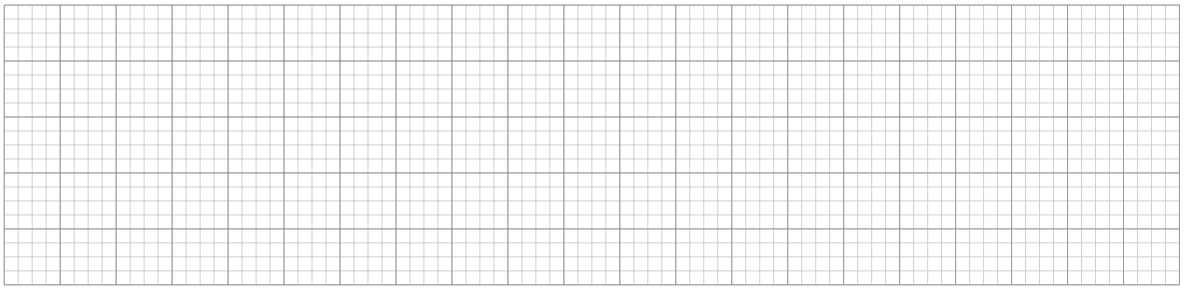
Property

Replacing each x_i by $kx_i + c$ gives a mean $k\overline{x} + c$, a standard deviation $|k|\sigma$ and a variance k^2V. Adding a constant changes neither variance nor standard deviation.

Example 7.10

Initial mean 12/20, standard deviation 3. Calculate the new statistics for three independent modifications, with no capping of marks: add 1.5; multiply by 0.8; multiply by 1.1 then add 0.5.

Solution:



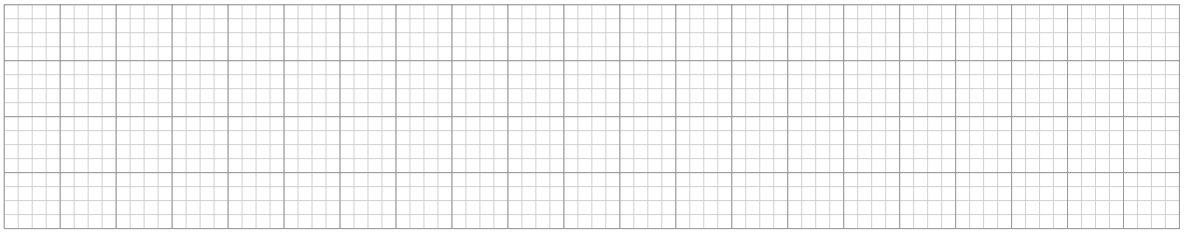
```
from math import sqrt

def moyenne_ecart_type(valeurs):
    n = len(valeurs)
    if n == 0:
        raise ValueError("liste vide")
    m = sum(valeurs)/n
    v = sum((x-m)**2 for x in valeurs)/n
    return m, sqrt(v)
```

Example 7.11

Explain the program; predict its result for `[8, 10, 12]`.

Solution:



VII Continuous data grouped into classes

Definitions and loss of information

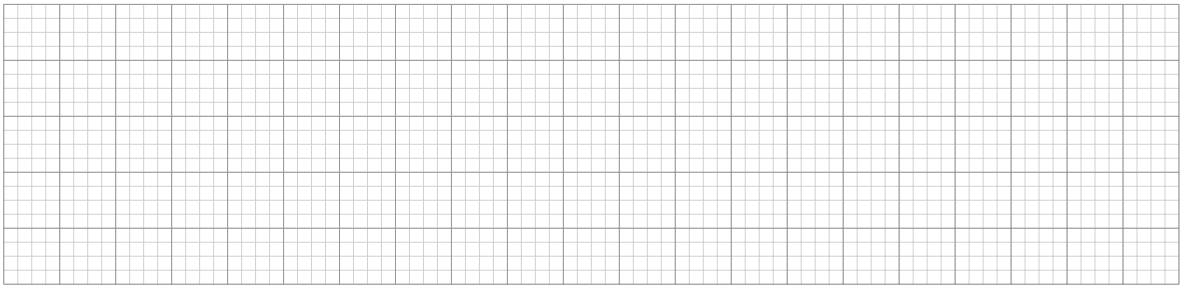
A continuous quantitative variable can take every value in an interval. Here, the observations are grouped into classes of equal width; the centre of $[a;b[$ is $(a+b)/2$. If the class means m_i are known, the exact overall mean is $\sum n_i m_i / N$. Otherwise, replacing them by the centres gives an estimate justified by a uniform model within each class.

Example 7.12: Heights – booklet data

Complete, estimate the mean from the centres and determine the median class.

Class (cm)	$[150;160[$	$[160;170[$	$[170;180[$	$[180;190[$	$[190;200[$
Count	8	22	35	28	7
Relative frequency (%)	.....	.....	.....	.....	.....
Centre (cm)	.....	.....	.....	.....	.....
ACC	.....	.....	.....	.....	.....

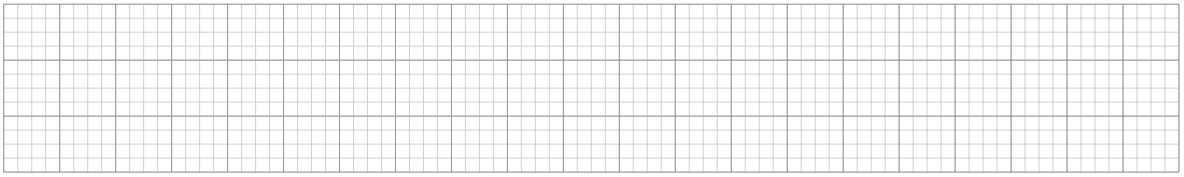
Solution:



Example 7.13

If the exact class means were $154;166;174;184;196$ cm, what would the overall mean be?

Solution:



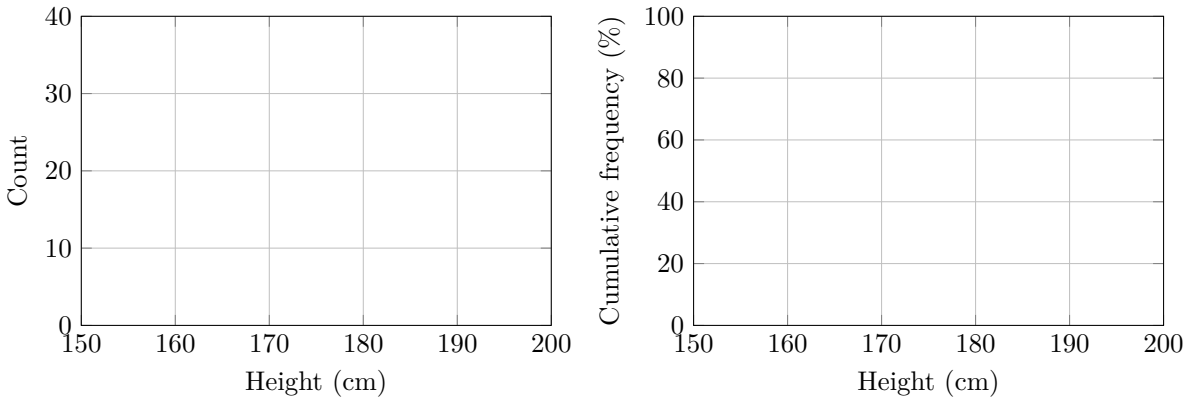
VIII Histogram and cumulative frequencies

Representations

The area of each rectangle of a histogram is proportional to the count. For equal widths, the heights can be the counts. The cumulative frequency polygon joins the points (upper bound; cumulative frequency), starting at the first lower bound with 0.

Example 7.14

Construct both graphs for the heights of the previous example.



Estimating the median

Under the assumption of uniformity in $[170; 180[$, the cumulative frequency rises linearly from 30 % to 65 %. Look for the abscissa corresponding to 50 %.

Example 7.15

Calculate this estimate and state its scope.

Solution: